

Assignment No	12
Title	Apriori and Clustering
Objective	Apriori , KMEANS, Agglomerative
Roll No	MCA2511

1. Apriori

apriori ample dataset.R

```
> #Install and load required packages  
> install.packages("arules")
```

Restarting R session...

```
> install.packages("arules")
```

WARNING: Rtools is required to build R packages but is not currently installed. Please download and install the appropriate version of Rtools before proceeding:

<https://cran.rstudio.com/bin/windows/Rtools/>

```
trying URL  
'https://cran.rstudio.com/bin/windows/contrib/4.5/arules_1.7-11.zip'  
Content type 'application/zip' length 2638349 bytes (2.5 MB)  
downloaded 2.5 MB
```

package 'arules' successfully unpacked and MD5 sums checked

Warning: cannot remove prior installation of package 'arules'
Warning: restored 'arules'

The downloaded binary packages are in
C:\Users\MCA2511\AppData\Local\Temp\RtmpGOpouW\downloaded_packages

Loading required namespace: arules

```
> install.packages("arulesViz")
```

WARNING: Rtools is required to build R packages but is not currently installed. Please download and install the appropriate version of Rtools before proceeding:

<https://cran.rstudio.com/bin/windows/Rtools/>

trying URL

'https://cran.rstudio.com/bin/windows/contrib/4.5/arulesViz_1.5.4.zip'

Content type 'application/zip' length 1661823 bytes (1.6 MB)
downloaded 1.6 MB

package 'arulesViz' successfully unpacked and MD5 sums checked

The downloaded binary packages are in

C:\Users\MCA2511\AppData\Local\Temp\RtmpGOpouW\downloaded_packages

Warning message:

In file.copy(savedcopy, lib, recursive = TRUE) :

problem copying

C:\Users\MCA2511\AppData\Local\Programs\R\R-4.5.2\library\00LOCK\arules\libs\x64\arules.dll to

C:\Users\MCA2511\AppData\Local\Programs\R\R-4.5.2\library\arules\libs\x64\arules.dll: Permission denied

> library(arules)

Loading required package: Matrix

Attaching package: 'arules'

The following objects are masked from 'package:base':

abbreviate, write

> library(arulesViz)

>

> data("Groceries")

> inspect(head(Groceries,3))

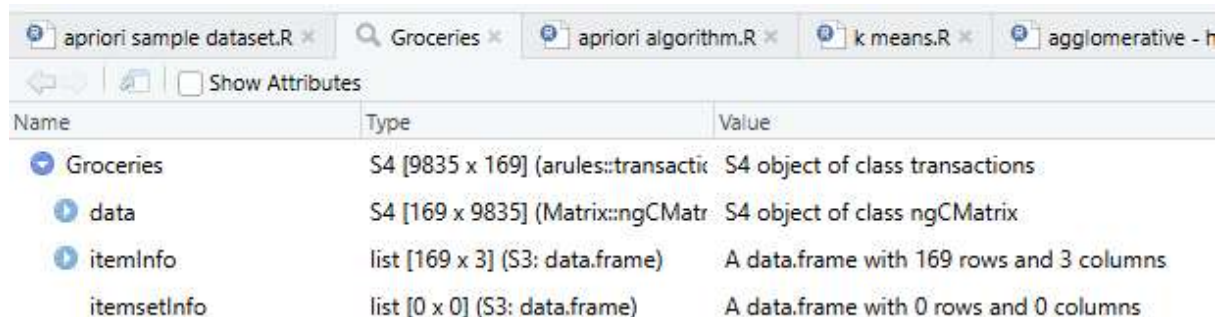
items

[1] {citrus fruit, semi-finished bread, margarine, ready soups}

[2] {tropical fruit, yogurt, coffee}

[3] {whole milk}

> View(Groceries)



Name	Type	Value
Groceries	S4 [9835 x 169] (arules::transacti	S4 object of class transactions
data	S4 [169 x 9835] (Matrix::ngCMatr	S4 object of class ngCMatrix
itemInfo	list [169 x 3] (S3: data.frame)	A data.frame with 169 rows and 3 columns
itemsetInfo	list [0 x 0] (S3: data.frame)	A data.frame with 0 rows and 0 columns

> rules <- apriori(Groceries,parameter =

```
+ list(support=0.01, confidence=0.09,maxlen=3,minlen=2))
Apriori
```

Parameter specification:

```
confidence minval smax arem aval originalSupport maxtime support minlen
maxlen target ext
0.09 0.1 1 none FALSE TRUE 5 0.01 2
3 rules TRUE
```

Algorithmic control:

```
filter tree heap memopt load sort verbose
0.1 TRUE TRUE FALSE TRUE 2 TRUE
```

Absolute minimum support count: 98

```
set item appearances ...[0 item(s)] done [0.00s].
set transactions ...[169 item(s), 9835 transaction(s)] done [0.00s].
sorting and recoding items ... [88 item(s)] done [0.00s].
creating transaction tree ... done [0.00s].
checking subsets of size 1 2 3 done [0.00s].
writing ... [445 rule(s)] done [0.00s].
creating S4 object ... done [0.00s].
```

Warning message:

```
In apriori(Groceries, parameter = list(support = 0.01, confidence = 0.09,
:
```

Mining stopped (maxlen reached). Only patterns up to a length of 3 returned!

```
> rules
```

```
set of 445 rules
```

```
> inspect(rules)
```

lhs	rhs	support
confidence coverage		
[1] {hard cheese}	=> {whole milk}	0.01006609
0.41078838 0.02450432		
[2] {butter milk}	=> {other vegetables}	0.01037112
0.37090909 0.02796136		
[3] {butter milk}	=> {whole milk}	0.01159126
0.41454545 0.02796136		
[4] {ham}	=> {whole milk}	0.01148958
0.44140625 0.02602949		
[5] {sliced cheese}	=> {whole milk}	0.01077783
0.43983402 0.02450432		
[6] {oil}	=> {whole milk}	0.01128622
0.40217391 0.02806304		
[7] {onions}	=> {other vegetables}	0.01423488
0.45901639 0.03101169		
[8] {onions}	=> {whole milk}	0.01209964
0.39016393 0.03101169		
[9] {berries}	=> {yogurt}	0.01057448
0.31804281 0.03324860		
[10] {berries}	=> {other vegetables}	0.01026945
0.30886850 0.03324860		
[11] {berries}	=> {whole milk}	0.01179461
0.35474006 0.03324860		

[12] {hamburger meat} 0.41590214 0.03324860	=> {other vegetables}	0.01382816
[13] {hamburger meat} 0.44342508 0.03324860	=> {whole milk}	0.01474326
[14] {hygiene articles} 0.38888889 0.03294357	=> {whole milk}	0.01281139
[15] {salty snack} 0.28494624 0.03782410	=> {other vegetables}	0.01077783
[16] {salty snack} 0.29569892 0.03782410	=> {whole milk}	0.01118454
[17] {sugar} 0.31831832 0.03385867	=> {other vegetables}	0.01077783
[18] {sugar} 0.44444444 0.03385867	=> {whole milk}	0.01504830
[19] {waffles} 0.26190476 0.03843416	=> {other vegetables}	0.01006609
[20] {waffles} 0.33068783 0.03843416	=> {whole milk}	0.01270971
[21] {long life bakery product} 0.28532609 0.03741739	=> {other vegetables}	0.01067616
[22] {long life bakery product} 0.36141304 0.03741739	=> {whole milk}	0.01352313
[23] {dessert} 0.31232877 0.03711235	=> {other vegetables}	0.01159126
[24] {dessert} 0.36986301 0.03711235	=> {whole milk}	0.01372649
[25] {canned beer} 0.14659686 0.07768175	=> {shopping bags}	0.01138790
[26] {shopping bags} 0.11558308 0.09852567	=> {canned beer}	0.01138790
[27] {canned beer} 0.17801047 0.07768175	=> {soda}	0.01382816
[28] {canned beer} 0.14528796 0.07768175	=> {rolls/buns}	0.01128622
[29] {cream cheese } 0.31282051 0.03965430	=> {yogurt}	0.01240468
[30] {cream cheese } 0.34615385 0.03965430	=> {other vegetables}	0.01372649
[31] {cream cheese } 0.41538462 0.03965430	=> {whole milk}	0.01647178
[32] {chicken} 0.25355450 0.04290798	=> {root vegetables}	0.01087951
[33] {root vegetables} 0.09981343 0.10899847	=> {chicken}	0.01087951
[34] {chicken} 0.41706161 0.04290798	=> {other vegetables}	0.01789527
[35] {other vegetables} 0.09248555 0.19349263	=> {chicken}	0.01789527
[36] {chicken} 0.40995261 0.04290798	=> {whole milk}	0.01759024
[37] {white bread} 0.24396135 0.04209456	=> {soda}	0.01026945
[38] {white bread} 0.32608696 0.04209456	=> {other vegetables}	0.01372649
[39] {white bread} 0.40579710 0.04209456	=> {whole milk}	0.01708185
[40] {chocolate} 0.27254098 0.04961871	=> {soda}	0.01352313

[41] {chocolate}	=> {rolls/buns}	0.01179461
0.23770492 0.04961871		
[42] {chocolate}	=> {other vegetables}	0.01270971
0.25614754 0.04961871		
[43] {chocolate}	=> {whole milk}	0.01667514
0.33606557 0.04961871		
[44] {coffee}	=> {rolls/buns}	0.01098119
0.18914186 0.05805796		
[45] {coffee}	=> {other vegetables}	0.01342145
0.23117338 0.05805796		
[46] {coffee}	=> {whole milk}	0.01870869
0.32224168 0.05805796		
[47] {frozen vegetables}	=> {root vegetables}	0.01159126
0.24101480 0.04809354		
[48] {root vegetables}	=> {frozen vegetables}	0.01159126
0.10634328 0.10899847		
[49] {frozen vegetables}	=> {yogurt}	0.01240468
0.25792812 0.04809354		
[50] {frozen vegetables}	=> {rolls/buns}	0.01016777
0.21141649 0.04809354		
[51] {frozen vegetables}	=> {other vegetables}	0.01779359
0.36997886 0.04809354		
[52] {other vegetables}	=> {frozen vegetables}	0.01779359
0.09196006 0.19349263		
[53] {frozen vegetables}	=> {whole milk}	0.02043721
0.42494715 0.04809354		
[54] {beef}	=> {root vegetables}	0.01738688
0.33139535 0.05246568		
[55] {root vegetables}	=> {beef}	0.01738688
0.15951493 0.10899847		
[56] {beef}	=> {yogurt}	0.01169293
0.22286822 0.05246568		
[57] {beef}	=> {rolls/buns}	0.01362481
0.25968992 0.05246568		
[58] {beef}	=> {other vegetables}	0.01972547
0.37596899 0.05246568		
[59] {other vegetables}	=> {beef}	0.01972547
0.10194430 0.19349263		
[60] {beef}	=> {whole milk}	0.02125064
0.40503876 0.05246568		
[61] {curd}	=> {whipped/sour cream}	0.01047280
0.19656489 0.05327911		
[62] {whipped/sour cream}	=> {curd}	0.01047280
0.14609929 0.07168277		
[63] {curd}	=> {tropical fruit}	0.01026945
0.19274809 0.05327911		
[64] {tropical fruit}	=> {curd}	0.01026945
0.09786822 0.10493137		
[65] {curd}	=> {root vegetables}	0.01087951
0.20419847 0.05327911		
[66] {root vegetables}	=> {curd}	0.01087951
0.09981343 0.10899847		
[67] {curd}	=> {yogurt}	0.01728521
0.32442748 0.05327911		
[68] {yogurt}	=> {curd}	0.01728521
0.12390671 0.13950178		
[69] {curd}	=> {rolls/buns}	0.01006609
0.18893130 0.05327911		

[70] {curd}
0.32251908 0.05327911
[71] {curd}
0.49045802 0.05327911
[72] {whole milk}
0.10226821 0.25551601
[73] {napkins}
0.19223301 0.05236401
[74] {tropical fruit}
0.09593023 0.10493137
[75] {napkins}
0.22912621 0.05236401
[76] {napkins}
0.23495146 0.05236401
[77] {napkins}
0.22330097 0.05236401
[78] {napkins}
0.27572816 0.05236401
[79] {napkins}
0.37669903 0.05236401
[80] {pork}
0.23633157 0.05765125
[81] {root vegetables}
0.12500000 0.10899847
[82] {pork}
0.20634921 0.05765125
[83] {pork}
0.19576720 0.05765125
[84] {pork}
0.37566138 0.05765125
[85] {other vegetables}
0.11192853 0.19349263
[86] {pork}
0.38447972 0.05765125
[87] {frankfurter}
0.17068966 0.05897306
[88] {sausage}
0.10714286 0.09395018
[89] {frankfurter}
0.17241379 0.05897306
[90] {root vegetables}
0.09328358 0.10899847
[91] {frankfurter}
0.19137931 0.05897306
[92] {frankfurter}
0.18965517 0.05897306
[93] {frankfurter}
0.32586207 0.05897306
[94] {rolls/buns}
0.10447761 0.18393493
[95] {frankfurter}
0.27931034 0.05897306
[96] {frankfurter}
0.34827586 0.05897306
[97] {bottled beer}
0.19570707 0.08052872
[98] {bottled water}
0.14259430 0.11052364

=> {other vegetables} 0.01718353
=> {whole milk} 0.02613116
=> {curd} 0.02613116
=> {tropical fruit} 0.01006609
=> {napkins} 0.01006609
=> {soda} 0.01199797
=> {yogurt} 0.01230300
=> {rolls/buns} 0.01169293
=> {other vegetables} 0.01443823
=> {whole milk} 0.01972547
=> {root vegetables} 0.01362481
=> {pork} 0.01362481
=> {soda} 0.01189629
=> {rolls/buns} 0.01128622
=> {other vegetables} 0.02165735
=> {pork} 0.02165735
=> {whole milk} 0.02216573
=> {sausage} 0.01006609
=> {frankfurter} 0.01006609
=> {root vegetables} 0.01016777
=> {frankfurter} 0.01016777
=> {soda} 0.01128622
=> {yogurt} 0.01118454
=> {rolls/buns} 0.01921708
=> {frankfurter} 0.01921708
=> {other vegetables} 0.01647178
=> {whole milk} 0.02053889
=> {bottled water} 0.01576004
=> {bottled beer} 0.01576004

[99] {bottled beer}	=> {soda}	0.01698017
0.21085859 0.08052872		
[100] {soda}	=> {bottled beer}	0.01698017
0.09737609 0.17437722		
[101] {bottled beer}	=> {rolls/buns}	0.01362481
0.16919192 0.08052872		
[102] {bottled beer}	=> {other vegetables}	0.01616675
0.20075758 0.08052872		
[103] {bottled beer}	=> {whole milk}	0.02043721
0.25378788 0.08052872		
[104] {brown bread}	=> {sausage}	0.01067616
0.16457680 0.06487036		
[105] {sausage}	=> {brown bread}	0.01067616
0.11363636 0.09395018		
[106] {brown bread}	=> {tropical fruit}	0.01067616
0.16457680 0.06487036		
[107] {tropical fruit}	=> {brown bread}	0.01067616
0.10174419 0.10493137		
[108] {brown bread}	=> {root vegetables}	0.01016777
0.15673981 0.06487036		
[109] {root vegetables}	=> {brown bread}	0.01016777
0.09328358 0.10899847		
[110] {brown bread}	=> {soda}	0.01260803
0.19435737 0.06487036		
[111] {brown bread}	=> {yogurt}	0.01453991
0.22413793 0.06487036		
[112] {yogurt}	=> {brown bread}	0.01453991
0.10422741 0.13950178		
[113] {brown bread}	=> {rolls/buns}	0.01260803
0.19435737 0.06487036		
[114] {brown bread}	=> {other vegetables}	0.01870869
0.28840125 0.06487036		
[115] {other vegetables}	=> {brown bread}	0.01870869
0.09668944 0.19349263		
[116] {brown bread}	=> {whole milk}	0.02521607
0.38871473 0.06487036		
[117] {whole milk}	=> {brown bread}	0.02521607
0.09868683 0.25551601		
[118] {margarine}	=> {bottled water}	0.01026945
0.17534722 0.05856634		
[119] {bottled water}	=> {margarine}	0.01026945
0.09291628 0.11052364		
[120] {margarine}	=> {root vegetables}	0.01108287
0.18923611 0.05856634		
[121] {root vegetables}	=> {margarine}	0.01108287
0.10167910 0.10899847		
[122] {margarine}	=> {soda}	0.01016777
0.17361111 0.05856634		
[123] {margarine}	=> {yogurt}	0.01423488
0.24305556 0.05856634		
[124] {yogurt}	=> {margarine}	0.01423488
0.10204082 0.13950178		
[125] {margarine}	=> {rolls/buns}	0.01474326
0.25173611 0.05856634		
lift count		
[1] 1.6076815 99		
[2] 1.9169159 102		
[3] 1.6223854 114		
[4] 1.7275091 113		

[5] 1.7213560 106
[6] 1.5739675 111
[7] 2.3722681 140
[8] 1.5269647 119
[9] 2.2798477 104
[10] 1.5962805 101
[11] 1.3883281 116
[12] 2.1494470 136
[13] 1.7354101 145
[14] 1.5219746 126
[15] 1.4726465 106
[16] 1.1572618 110
[17] 1.6451186 106
[18] 1.7393996 148
[19] 1.3535645 99
[20] 1.2941961 125
[21] 1.4746096 105
[22] 1.4144438 133
[23] 1.6141636 114
[24] 1.4475140 135
[25] 1.4879052 112
[26] 1.4879052 112
[27] 1.0208356 136
[28] 0.7898878 111
[29] 2.2424123 122
[30] 1.7889769 135
[31] 1.6256696 162
[32] 2.3262206 107
[33] 2.3262206 107
[34] 2.1554393 176
[35] 2.1554393 176
[36] 1.6044106 173
[37] 1.3990437 101
[38] 1.6852681 135
[39] 1.5881474 168
[40] 1.5629391 133
[41] 1.2923316 116
[42] 1.3238103 125
[43] 1.3152427 164
[44] 1.0283085 108
[45] 1.1947400 132
[46] 1.2611408 184
[47] 2.2111759 114
[48] 2.2111759 114
[49] 1.8489235 122
[50] 1.1494092 100
[51] 1.9121083 175
[52] 1.9121083 175
[53] 1.6630940 201
[54] 3.0403668 171
[55] 3.0403668 171
[56] 1.5976012 115
[57] 1.4118576 134
[58] 1.9430662 194
[59] 1.9430662 194
[60] 1.5851795 209
[61] 2.7421499 103
[62] 2.7421499 103
[63] 1.8368968 101

[64] 1.8368968 101
[65] 1.8734067 107
[66] 1.8734067 107
[67] 2.3256154 170
[68] 2.3256154 170
[69] 1.0271638 99
[70] 1.6668288 169
[71] 1.9194805 257
[72] 1.9194805 257
[73] 1.8319880 99
[74] 1.8319880 99
[75] 1.3139687 118
[76] 1.6842183 121
[77] 1.2140216 115
[78] 1.4250060 142
[79] 1.4742678 194
[80] 2.1682099 134
[81] 2.1682099 134
[82] 1.1833495 117
[83] 1.0643286 111
[84] 1.9414764 213
[85] 1.9414764 213
[86] 1.5047187 218
[87] 1.8168103 99
[88] 1.8168103 99
[89] 1.5818001 100
[90] 1.5818001 100
[91] 1.0975018 111
[92] 1.3595179 110
[93] 1.7716161 189
[94] 1.7716161 189
[95] 1.4435193 162
[96] 1.3630295 202
[97] 1.7707259 155
[98] 1.7707259 155
[99] 1.2092094 167
[100] 1.2092094 167
[101] 0.9198466 134
[102] 1.0375464 159
[103] 0.9932367 201
[104] 1.7517455 105
[105] 1.7517455 105
[106] 1.5684233 105
[107] 1.5684233 105
[108] 1.4380000 100
[109] 1.4380000 100
[110] 1.1145800 124
[111] 1.6067030 143
[112] 1.6067030 143
[113] 1.0566637 124
[114] 1.4905025 184
[115] 1.4905025 184
[116] 1.5212930 248
[117] 1.5212930 248
[118] 1.5865133 101
[119] 1.5865133 101
[120] 1.7361354 109
[121] 1.7361354 109
[122] 0.9956066 100

```
[123] 1.7423115 140  
[124] 1.7423115 140  
[125] 1.3686151 145  
[ reached 'max' / getOption("max.print") -- omitted 320 rows ]
```

apriori algorithm.R

```
> #Install and load required packages  
> install.packages("arules")
```

Restarting R session...

```
> install.packages("arules")
```

WARNING: Rtools is required to build R packages but is not currently installed. Please download and install the appropriate version of Rtools before proceeding:

<https://cran.rstudio.com/bin/windows/Rtools/>

```
trying URL  
'https://cran.rstudio.com/bin/windows/contrib/4.5/arules_1.7-11.zip'  
Content type 'application/zip' length 2638349 bytes (2.5 MB)  
downloaded 2.5 MB
```

package 'arules' successfully unpacked and MD5 sums checked

Warning: cannot remove prior installation of package 'arules'

Warning: restored 'arules'

The downloaded binary packages are in
C:\Users\MCA2511\AppData\Local\Temp\Rtmpy4aqMx\downloaded_packages

Loading required namespace: arules

```
> install.packages("arulesViz")
```

WARNING: Rtools is required to build R packages but is not currently installed. Please download and install the appropriate version of Rtools before proceeding:

<https://cran.rstudio.com/bin/windows/Rtools/>

```
trying URL  
'https://cran.rstudio.com/bin/windows/contrib/4.5/arulesViz_1.5.4.zip'  
Content type 'application/zip' length 1661823 bytes (1.6 MB)  
downloaded 1.6 MB
```

package 'arulesViz' successfully unpacked and MD5 sums checked

The downloaded binary packages are in
C:\Users\MCA2511\AppData\Local\Temp\Rtmpy4aqMx\downloaded_packages

Warning message:

In file.copy(savedcopy, lib, recursive = TRUE) :

problem copying

```
C:\Users\MCA2511\AppData\Local\Programs\R\R-4.5.2\library\00LOCK\arules\libs\x64\arules.dll to
```

```
C:\Users\MCA2511\AppData\Local\Programs\R\R-4.5.2\library\arules\libs\x64\arules.dll: Permission denied
```

```
>
```

```
> library(arules)
```

Loading required package: Matrix

Attaching package: 'arules'

The following objects are masked from 'package:base':

abbreviate, write

```
> library(arulesViz)
```

```
> getwd()
```

```
[1] "A:/MCA2511/ADBMS_Lab/R"
```

```
> setwd("A:/MCA2511/ADBMS_Lab/R")
```

```
>
```

```
> #Read CSV file
```

```
> #Make sure data_apriori.csv saved in Working Directory
```

```
> mba_data <- read.csv("data_apriori.csv", stringsAsFactors = FALSE)
```

```
> #Check the structure
```

```
> str(mba_data)
```

```
'data.frame':    50 obs. of  2 variables:
```

```
 $ Customer_Id: int  1 2 3 4 5 6 7 8 9 10 ...
```

```
 $ Products    : chr  "laptop, mouse , headphones, pendrive, speakers"
```

```
"laptop, headphones" "laptop, mouse, pendrive" "mouse, speakers" ...
```

```
> View(mba_data)
```

Filter	
Customer_id	Products
1	laptop, mouse , headphones, pendrive, speakers
2	laptop, headphones
3	laptop, mouse, pendrive
4	mouse, speakers
5	laptop, pendrive
6	laptop, mouse , headphones, pendrive, speakers
7	laptop, headphones
8	laptop, mouse, pendrive
9	mouse, speakers
10	laptop, pendrive
11	laptop, mouse , headphones, pendrive, speakers
12	laptop, headphones
13	laptop, mouse, pendrive

```
> #Ensure correct column names(case-sensitive)
> colnames(mba_data) <- c("Customer_id","Products")
> #Split 'Products' into individual items
> trans_list <- strsplit(mba_data$Products, ",") #Split by comma
> trans_list <- lapply(trans_list, trimws) #Remove leading/trailing spaces
> #Assign customer IDs as transactional names
> names(trans_list) <- mba_data$Customer_id
> #Convert list to 'transactions' object
> trans <- as(trans_list,"transactions")
> #Inspect the transactions
> inspect(head(trans,5))
```

```
      items                                     transactionID
[1] {headphones, laptop, mouse, pendrive, speakers} 1
[2] {headphones, laptop}                             2
[3] {laptop, mouse, pendrive}                         3
[4] {mouse, speakers}                                4
[5] {laptop, pendrive}                                5
> itemLabels(trans)
[1] "headphones" "laptop"      "mouse"      "pendrive"   "speakers"
> summary(trans)
transactions as itemMatrix in sparse format with
  50 rows (elements/itemsets/transactions) and
  5 columns (items) and a density of 0.588
```

most frequent items:

laptop	mouse	pendrive	headphones	speakers	(Other)
42	31	31	23	20	0

element (itemset/transaction) length distribution:

```
sizes
 2  3  5
27 11 12
```

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
2.00	2.00	2.00	2.94	3.00	5.00

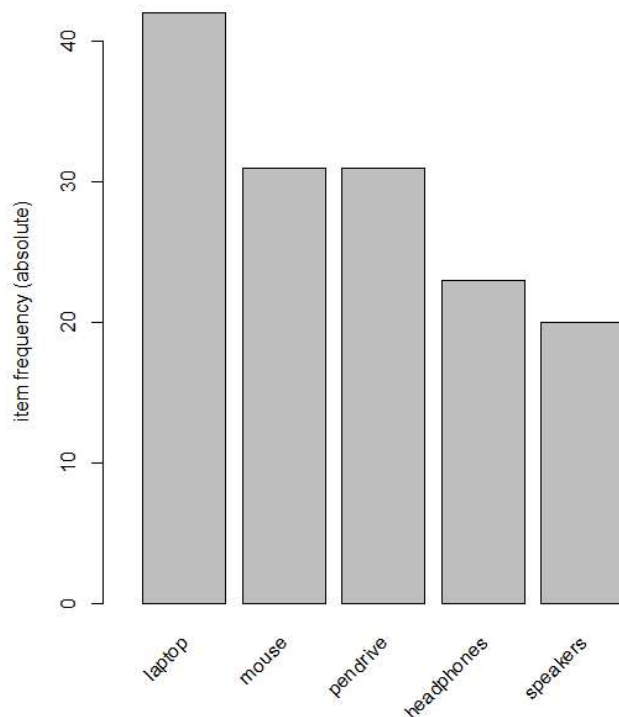
includes extended item information - examples:

```
      labels
1 headphones
2      laptop
3        mouse
```

includes extended transaction information - examples:

```
      transactionID
1                1
2                2
3                3
> #Plot items frequencies
> itemFrequencyPlot(trans, topN=10,type="absolute",main="Top 10 most
frequent Itemset")
```

Top 10 most frequent itemset



```
> #Generate Apriori rules
> rules <- apriori(
+ trans,
+ parameter = list(support = 0.05, confidence = 0.3, minlen = 2, maxlen =
3)
+ )
```

Apriori

Parameter specification:

```
confidence minval smax arem aval originalSupport maxtime support minlen
0.3 0.1 1 none FALSE TRUE 5 0.05 2
maxlen target ext
3 rules TRUE
```

Algorithmic control:

```
filter tree heap memopt load sort verbose
0.1 TRUE TRUE FALSE TRUE 2 TRUE
```

Absolute minimum support count: 2

```
set item appearances ...[0 item(s)] done [0.00s].
set transactions ...[5 item(s), 50 transaction(s)] done [0.00s].
sorting and recoding items ... [5 item(s)] done [0.00s].
creating transaction tree ... done [0.00s].
checking subsets of size 1 2 3 done [0.00s].
writing ... [49 rule(s)] done [0.00s].
creating S4 object ... done [0.00s].
```

Warning message:

In apriori(trans, parameter = list(support = 0.05, confidence = 0.3, :
Mining stopped (maxlen reached). Only patterns up to a length of 3
returned!

> #Check rule Summary

> summary(rules)

set of 49 rules

rule length distribution (lhs + rhs):sizes

2 3

19 30

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
2.000	2.000	3.000	2.612	3.000	3.000

summary of quality measures:

support	confidence	coverage	lift
Min. :0.2400	Min. :0.3871	Min. :0.2400	Min. :0.7143
1st Qu.:0.2400	1st Qu.:0.5217	1st Qu.:0.2400	1st Qu.:0.9677
Median :0.2400	Median :0.7381	Median :0.4600	Median :1.1905
Mean :0.3024	Mean :0.7388	Mean :0.4433	Mean :1.2823
3rd Qu.:0.4000	3rd Qu.:1.0000	3rd Qu.:0.6200	3rd Qu.:1.6129
Max. :0.6200	Max. :1.0000	Max. :0.8400	Max. :2.5000

count
Min. :12.00
1st Qu.:12.00
Median :12.00
Mean :15.12
3rd Qu.:20.00
Max. :31.00

mining info:

data	ntransactions	support	confidence
trans	50	0.05	0.3

call

apriori(data = trans, parameter = list(support = 0.05, confidence = 0.3,
minlen = 2, maxlen = 3))

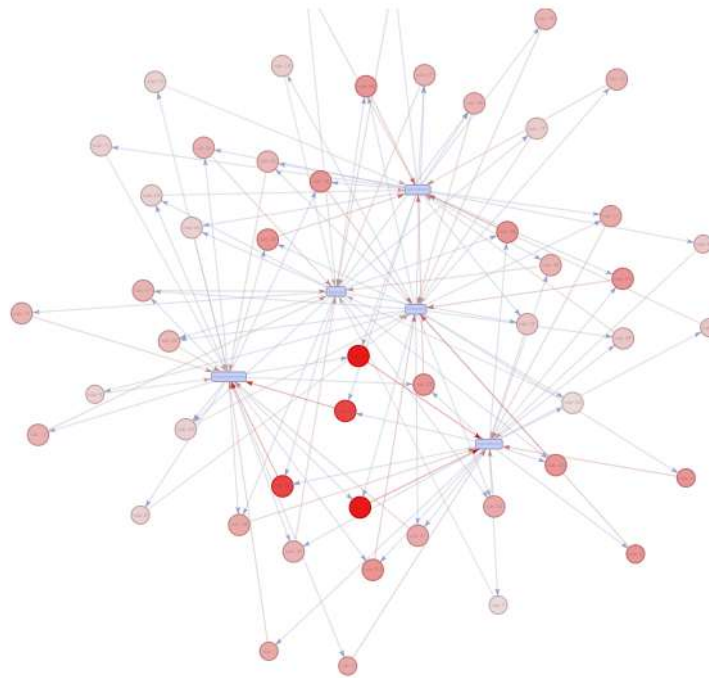
> #Inspect top rules by lift

> inspect(head(sort(rules, by = "lift"),10))

	lhs	rhs	support	confidence	coverage
[1]	{headphones, mouse}	=> {speakers}	0.24	1.0000000	0.24
[2]	{headphones, pendrive}	=> {speakers}	0.24	1.0000000	0.24
[3]	{pendrive, speakers}	=> {headphones}	0.24	1.0000000	0.24
[4]	{laptop, speakers}	=> {headphones}	0.24	1.0000000	0.24
[5]	{speakers}	=> {mouse}	0.40	1.0000000	0.40
[6]	{mouse}	=> {speakers}	0.40	0.6451613	0.62
[7]	{headphones, speakers}	=> {mouse}	0.24	1.0000000	0.24

```
[8] {headphones, speakers} => {pendrive}    0.24    1.00000000  0.24
[9] {pendrive, speakers}   => {mouse}       0.24    1.00000000  0.24
[10] {laptop, speakers}     => {mouse}       0.24    1.00000000  0.24
      lift      count
[1]  2.500000  12
[2]  2.500000  12
[3]  2.173913  12
[4]  2.173913  12
[5]  1.612903  20
[6]  1.612903  20
[7]  1.612903  12
[8]  1.612903  12
[9]  1.612903  12
[10] 1.612903  12
> # Visualize rules (interactive graph)
> plot(rules,method = "graph", engine = "htmlwidget")
```

Select by id ▼



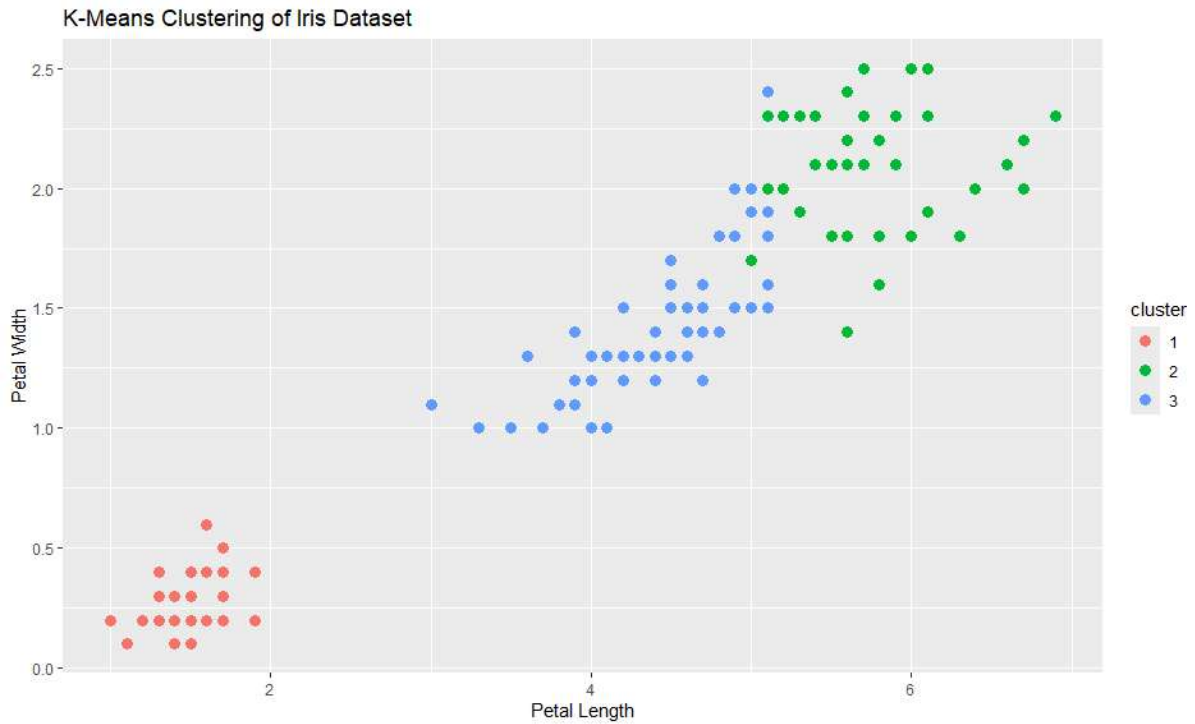
```
> #Optional: Export rules to CSV
> rules_df <- as(rules,"data.frame")
> write.csv(rules_df,"apriori_rules_output.csv", row.names = FALSE)
> #Print final confirmation
> cat("Apriori analysis completed Rules saved to 'apriori rules
output.csv'\n")
Apriori analysis completed Rules saved to 'apriori rules output.csv'
```


Rules:

	A	B	C	D	E	F
1	rules	support	confidence	coverage	lift	count
2	{speakers} => {headphones}	0.24	0.6	0.4	1.304347826	12
3	{headphones} => {speakers}	0.24	0.52173913	0.46	1.304347826	12
4	{speakers} => {mouse}	0.4	1	0.4	1.612903226	20
5	{mouse} => {speakers}	0.4	0.64516129	0.62	1.612903226	20
6	{speakers} => {pendrive}	0.24	0.6	0.4	0.967741935	12
7	{pendrive} => {speakers}	0.24	0.387096774	0.62	0.967741935	12
8	{speakers} => {laptop}	0.24	0.6	0.4	0.714285714	12
9	{headphones} => {mouse}	0.24	0.52173913	0.46	0.841514727	12
10	{mouse} => {headphones}	0.24	0.387096774	0.62	0.841514727	12
11	{headphones} => {pendrive}	0.24	0.52173913	0.46	0.841514727	12
12	{pendrive} => {headphones}	0.24	0.387096774	0.62	0.841514727	12
13	{headphones} => {laptop}	0.46	1	0.46	1.19047619	23
14	{laptop} => {headphones}	0.46	0.547619048	0.84	1.19047619	23
15	{mouse} => {pendrive}	0.46	0.741935484	0.62	1.196670135	23
16	{pendrive} => {mouse}	0.46	0.741935484	0.62	1.196670135	23
17	{mouse} => {laptop}	0.46	0.741935484	0.62	0.883256528	23
18	{laptop} => {mouse}	0.46	0.547619048	0.84	0.883256528	23
19	{pendrive} => {laptop}	0.62	1	0.62	1.19047619	31
20	{laptop} => {pendrive}	0.62	0.738095238	0.84	1.19047619	31
21	{headphones,speakers} => {mouse}	0.24	1	0.24	1.612903226	12

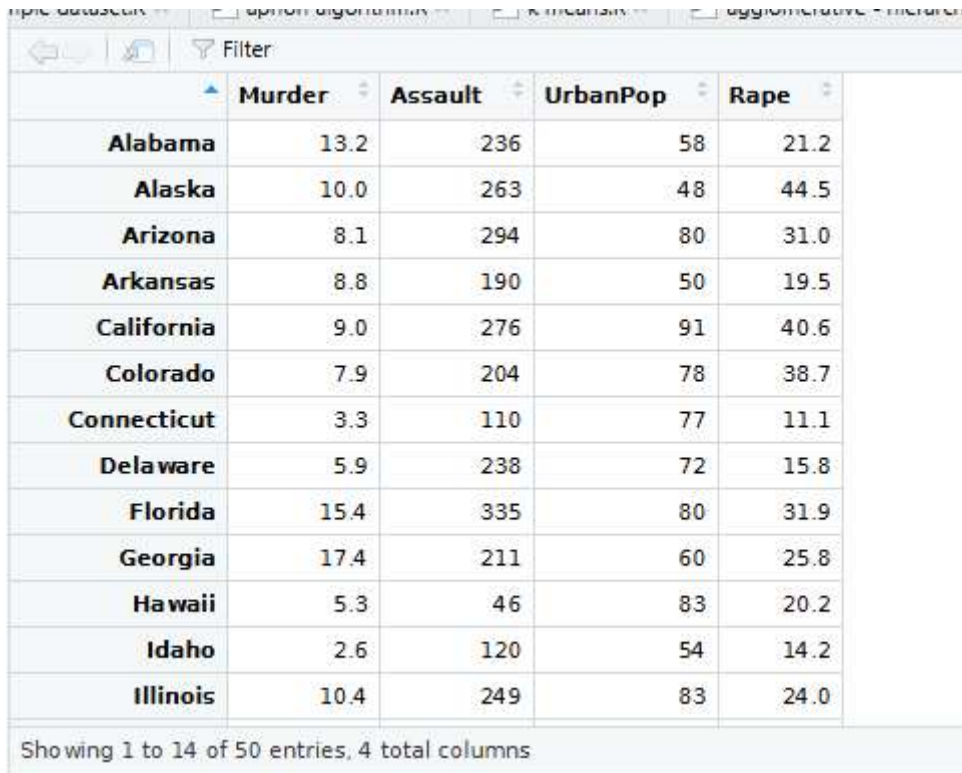
```
> ggplot(iris, aes(Petal.Length, Petal.Width, color = cluster)) +
```

```
+ geom_point(size = 3) +  
+ labs(title = "K-Means Clustering of Iris Dataset",  
+       x = "Petal Length",  
+       y = "Petal Width")
```



3. Agglomerative

```
> df <- USArrests
> View(df) #Opens the dataset in a spreadsheet-style viewer
```



	Murder	Assault	UrbanPop	Rape
Alabama	13.2	236	58	21.2
Alaska	10.0	263	48	44.5
Arizona	8.1	294	80	31.0
Arkansas	8.8	190	50	19.5
California	9.0	276	91	40.6
Colorado	7.9	204	78	38.7
Connecticut	3.3	110	77	11.1
Delaware	5.9	238	72	15.8
Florida	15.4	335	80	31.9
Georgia	17.4	211	60	25.8
Hawaii	5.3	46	83	20.2
Idaho	2.6	120	54	14.2
Illinois	10.4	249	83	24.0

Showing 1 to 14 of 50 entries, 4 total columns

```
> df <- na.omit(df)
> df
```

	Murder	Assault	UrbanPop	Rape
Alabama	13.2	236	58	21.2
Alaska	10.0	263	48	44.5
Arizona	8.1	294	80	31.0
Arkansas	8.8	190	50	19.5
California	9.0	276	91	40.6
Colorado	7.9	204	78	38.7
Connecticut	3.3	110	77	11.1
Delaware	5.9	238	72	15.8
Florida	15.4	335	80	31.9
Georgia	17.4	211	60	25.8
Hawaii	5.3	46	83	20.2
Idaho	2.6	120	54	14.2
Illinois	10.4	249	83	24.0
Indiana	7.2	113	65	21.0
Iowa	2.2	56	57	11.3
Kansas	6.0	115	66	18.0
Kentucky	9.7	109	52	16.3
Louisiana	15.4	249	66	22.2
Maine	2.1	83	51	7.8

Maryland	11.3	300	67	27.8
Massachusetts	4.4	149	85	16.3
Michigan	12.1	255	74	35.1
Minnesota	2.7	72	66	14.9
Mississippi	16.1	259	44	17.1
Missouri	9.0	178	70	28.2
Montana	6.0	109	53	16.4
Nebraska	4.3	102	62	16.5
Nevada	12.2	252	81	46.0
New Hampshire	2.1	57	56	9.5
New Jersey	7.4	159	89	18.8
New Mexico	11.4	285	70	32.1
New York	11.1	254	86	26.1
North Carolina	13.0	337	45	16.1
North Dakota	0.8	45	44	7.3
Ohio	7.3	120	75	21.4
Oklahoma	6.6	151	68	20.0
Oregon	4.9	159	67	29.3
Pennsylvania	6.3	106	72	14.9
Rhode Island	3.4	174	87	8.3
South Carolina	14.4	279	48	22.5
South Dakota	3.8	86	45	12.8
Tennessee	13.2	188	59	26.9
Texas	12.7	201	80	25.5
Utah	3.2	120	80	22.9
Vermont	2.2	48	32	11.2
Virginia	8.5	156	63	20.7
Washington	4.0	145	73	26.2
West Virginia	5.7	81	39	9.3
Wisconsin	2.6	53	66	10.8
Wyoming	6.8	161	60	15.6

```
> d <- scale(df)
```

```
> head(d) #Shows first few standardized rows
```

	Murder	Assault	UrbanPop	Rape
Alabama	1.24256408	0.7828393	-0.5209066	-0.003416473
Alaska	0.50786248	1.1068225	-1.2117642	2.484202941
Arizona	0.07163341	1.4788032	0.9989801	1.042878388
Arkansas	0.23234938	0.2308680	-1.0735927	-0.184916602
California	0.27826823	1.2628144	1.7589234	2.067820292
Colorado	0.02571456	0.3988593	0.8608085	1.864967207

```
> dist_matrix <- dist(d, method = "euclidean")
```

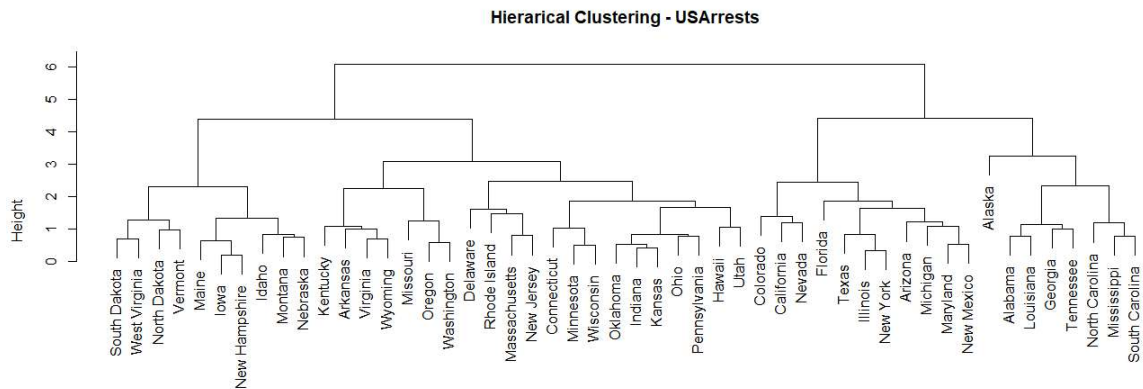
```
> hc <- hclust(dist_matrix, method = "complete")
```

```
> plot(hc,
```

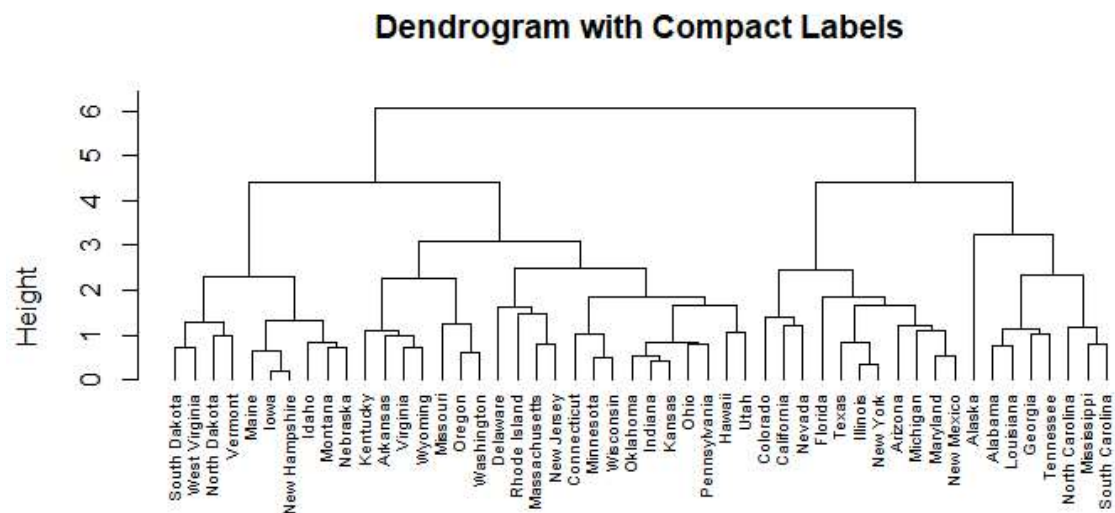
```
+   main = "Hierarical Clustering - USArrests",
```

```
+   xlab = "",
```

```
+   sub = "")
```



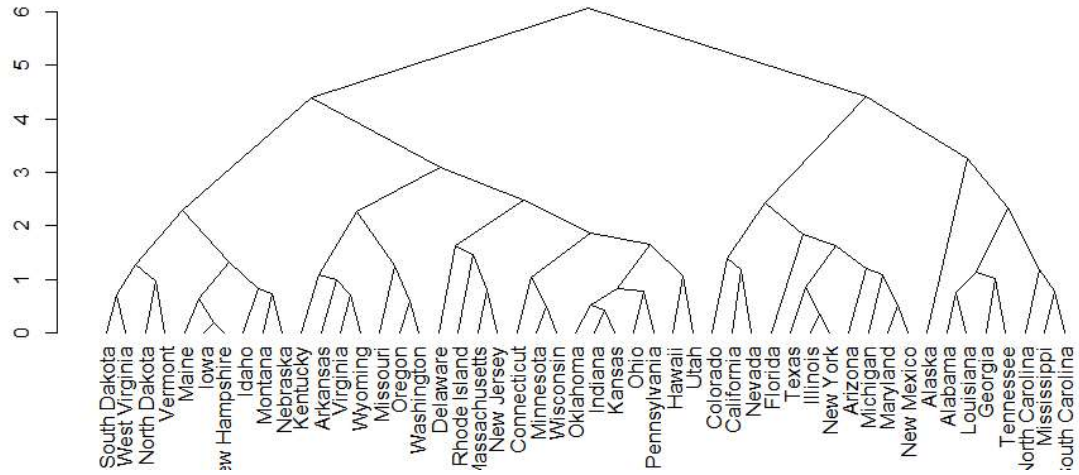
```
> plot(hc,
+       cex = 0.6,
+       hang = -1,
+       main = "Dendrogram with Compact Labels")
```



dist_matrix
hclust (*, "complete")

```
> hcd <- as.dendrogram(hc)
> plot(hcd,
+       type = "triangle",
+       main = "Triangular Dendrogram - USArrests")
```

Triangular Dendrogram - USArrests



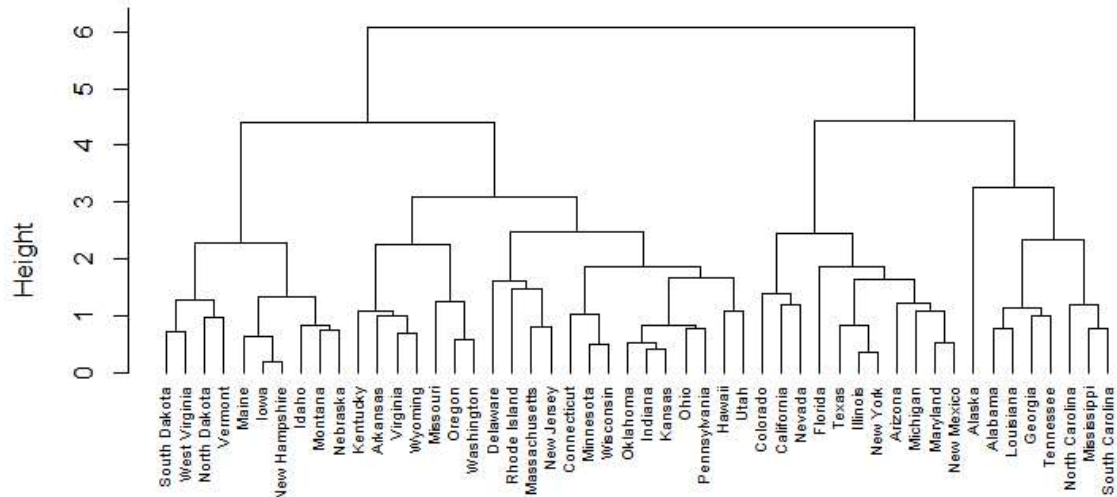
```
> groups <- cutree(hc, k=4)
> df$cluster <- groups
> head(df)
```

	Murder	Assault	UrbanPop	Rape
Alabama	13.2	236	58	21.2
Alaska	10.0	263	48	44.5
Arizona	8.1	294	80	31.0
Arkansas	8.8	190	50	19.5
California	9.0	276	91	40.6
Colorado	7.9	204	78	38.7

	cluster
Alabama	1
Alaska	1
Arizona	2
Arkansas	3
California	2
Colorado	2

```
> plot(hc, cex = 0.6, hang = -1)
```

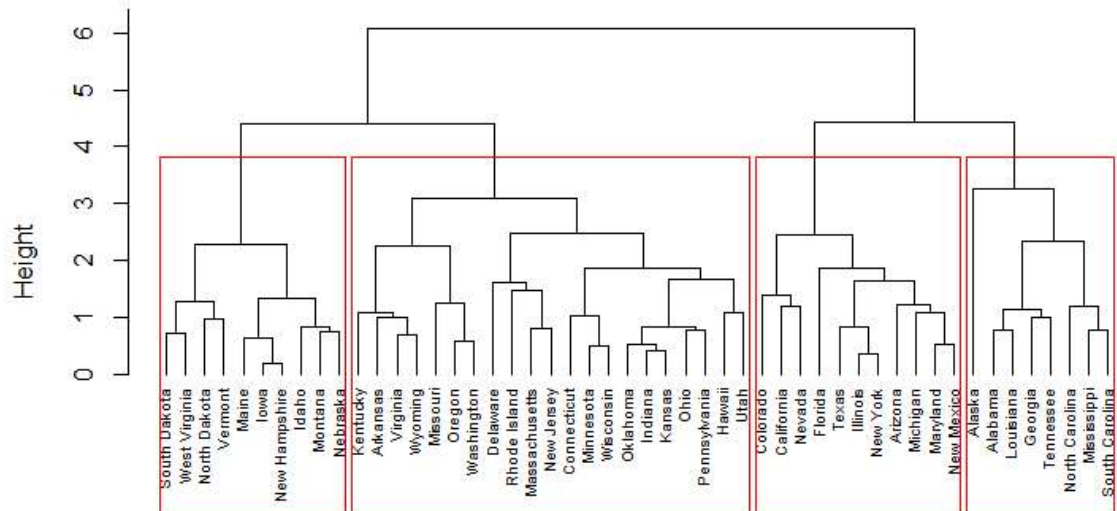

Cluster Dendrogram



```
dist_matrix
hclust(*, "complete")
```

```
> rect.hclust(hc, k = 4, border = "red")
```

Cluster Dendrogram



```
dist_matrix
hclust(*, "complete")
```

```
> #Cut the tree at height 75
> cut_tree <- cut(hcd, h = 75)
> #View Structure
> str(cut_tree)
```


List of 2

```
$ upper: ..--[dendrogram w/ 2 branches and 2 members at h = 6.08,
midpoint = 24.7, x.member = 50]
.. |--leaf "Branch 1" (h= 4.4 midpoint = 10.5, x.member = 31 )
.. `--leaf "Branch 2" (h= 4.42 midpoint = 7.88, x.member = 19 )
$ lower:List of 2
..$ : .. ..--[dendrogram w/ 2 branches and 31 members at h = 4.4,
midpoint = 10.5]
.. .. |--[dendrogram w/ 2 branches and 10 members at h = 2.3, midpoint =
3.88]
.. .. | |--[dendrogram w/ 2 branches and 4 members at h = 1.27,
midpoint = 1.5]
.. .. | | |--[dendrogram w/ 2 branches and 2 members at h = 0.711,
midpoint = 0.5]
.. .. | | | |--leaf "South Dakota"
.. .. | | | `--leaf "West Virginia"
.. .. | | `--[dendrogram w/ 2 branches and 2 members at h = 0.982,
midpoint = 0.5]
.. .. | | | |--leaf "North Dakota"
.. .. | | | `--leaf "Vermont"
.. .. | `--[dendrogram w/ 2 branches and 6 members at h = 1.33,
midpoint = 2.25]
.. .. | | |--[dendrogram w/ 2 branches and 3 members at h = 0.646,
midpoint = 0.75]
.. .. | | | |--leaf "Maine"
.. .. | | | `--[dendrogram w/ 2 branches and 2 members at h = 0.206,
midpoint = 0.5]
.. .. | | | | |--leaf "Iowa"
.. .. | | | | `--leaf "New Hampshire"
.. .. | | | `--[dendrogram w/ 2 branches and 3 members at h = 0.829,
midpoint = 0.75]
.. .. | | | | |--leaf "Idaho"
.. .. | | | | `--[dendrogram w/ 2 branches and 2 members at h = 0.739,
midpoint = 0.5]
.. .. | | | | | |--leaf "Montana"
.. .. | | | | | `--leaf "Nebraska"
.. .. | | | `--[dendrogram w/ 2 branches and 21 members at h = 3.09, midpoint
= 7.07]
.. .. | | |--[dendrogram w/ 2 branches and 7 members at h = 2.26,
midpoint = 2.81]
.. .. | | | |--[dendrogram w/ 2 branches and 4 members at h = 1.09,
midpoint = 0.875]
.. .. | | | | |--leaf "Kentucky"
.. .. | | | | `--[dendrogram w/ 2 branches and 3 members at h = 0.997,
midpoint = 0.75]
.. .. | | | | | |--leaf "Arkansas"
.. .. | | | | | `--[dendrogram w/ 2 branches and 2 members at h =
0.704, midpoint = 0.5]
.. .. | | | | | | |--leaf "Virginia"
.. .. | | | | | | `--leaf "Wyoming"
.. .. | | | | `--[dendrogram w/ 2 branches and 3 members at h = 1.25,
```

```

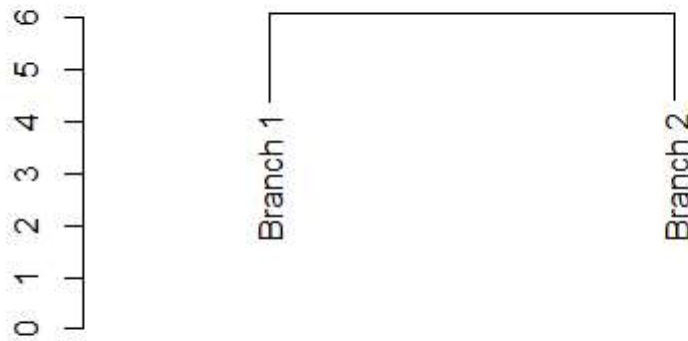
midpoint = 0.75]
.. .. | |--leaf "Missouri"
.. .. | `--[dendrogram w/ 2 branches and 2 members at h = 0.594,
midpoint = 0.5]
.. .. | |--leaf "Oregon"
.. .. | `--leaf "Washington"
.. .. `--[dendrogram w/ 2 branches and 14 members at h = 2.47,
midpoint = 4.33]
.. .. |--[dendrogram w/ 2 branches and 4 members at h = 1.62,
midpoint = 0.875]
.. .. | |--leaf "Delaware"
.. .. | `--[dendrogram w/ 2 branches and 3 members at h = 1.47,
midpoint = 0.75]
.. .. | |--leaf "Rhode Island"
.. .. | `--[dendrogram w/ 2 branches and 2 members at h =
0.798, midpoint = 0.5]
.. .. | |--leaf "Massachusetts"
.. .. | `--leaf "New Jersey"
.. .. `--[dendrogram w/ 2 branches and 10 members at h = 1.86,
midpoint = 3.78]
.. .. |--[dendrogram w/ 2 branches and 3 members at h = 1.04,
midpoint = 0.75]
.. .. | |--leaf "Connecticut"
.. .. | `--[dendrogram w/ 2 branches and 2 members at h =
0.494, midpoint = 0.5]
.. .. | |--leaf "Minnesota"
.. .. | `--leaf "Wisconsin"
.. .. `--[dendrogram w/ 2 branches and 7 members at h = 1.66,
midpoint = 3.81]
.. .. |--[dendrogram w/ 2 branches and 5 members at h =
0.841, midpoint = 2.12]
.. .. | |--[dendrogram w/ 2 branches and 3 members at h =
0.53, midpoint = 0.75]
.. .. | | |--leaf "Oklahoma"
.. .. | | `--[dendrogram w/ 2 branches and 2 members at h =
= 0.429, midpoint = 0.5]
.. .. | | |--leaf "Indiana"
.. .. | | `--leaf "Kansas"
.. .. | `--[dendrogram w/ 2 branches and 2 members at h =
0.778, midpoint = 0.5]
.. .. | |--leaf "Ohio"
.. .. | `--leaf "Pennsylvania"
.. .. `--[dendrogram w/ 2 branches and 2 members at h =
1.07, midpoint = 0.5]
.. .. |--leaf "Hawaii"
.. .. `--leaf "Utah"
..$ : .. ..--[dendrogram w/ 2 branches and 19 members at h = 4.42,
midpoint = 7.88]
.. .. |--[dendrogram w/ 2 branches and 11 members at h = 2.45, midpoint
= 2.7]
.. .. | |--[dendrogram w/ 2 branches and 3 members at h = 1.4, midpoint

```

```
= 0.75]
.. .. | | |--leaf "Colorado"
.. .. | | `--[dendrogram w/ 2 branches and 2 members at h = 1.2,
midpoint = 0.5]
.. .. | | |--leaf "California"
.. .. | | `--leaf "Nevada"
.. .. | `--[dendrogram w/ 2 branches and 8 members at h = 1.85,
midpoint = 1.66]
.. .. | |--leaf "Florida"
.. .. | `--[dendrogram w/ 2 branches and 7 members at h = 1.64,
midpoint = 2.31]
.. .. | |--[dendrogram w/ 2 branches and 3 members at h = 0.846,
midpoint = 0.75]
.. .. | | |--leaf "Texas"
.. .. | | `--[dendrogram w/ 2 branches and 2 members at h =
0.35, midpoint = 0.5]
.. .. | | |--leaf "Illinois"
.. .. | | `--leaf "New York"
.. .. | `--[dendrogram w/ 2 branches and 4 members at h = 1.21,
midpoint = 0.875]
.. .. | |--leaf "Arizona"
.. .. | `--[dendrogram w/ 2 branches and 3 members at h =
1.08, midpoint = 0.75]
.. .. | |--leaf "Michigan"
.. .. | `--[dendrogram w/ 2 branches and 2 members at h =
0.535, midpoint = 0.5]
.. .. | |--leaf "Maryland"
.. .. | `--leaf "New Mexico"
.. .. `--[dendrogram w/ 2 branches and 8 members at h = 3.26, midpoint =
2.06]
.. .. |--leaf "Alaska"
.. .. `--[dendrogram w/ 2 branches and 7 members at h = 2.34,
midpoint = 3.12]
.. .. |--[dendrogram w/ 2 branches and 4 members at h = 1.13,
midpoint = 1.5]
.. .. | |--[dendrogram w/ 2 branches and 2 members at h = 0.772,
midpoint = 0.5]
.. .. | | |--leaf "Alabama"
.. .. | | `--leaf "Louisiana"
.. .. | `--[dendrogram w/ 2 branches and 2 members at h = 1.01,
midpoint = 0.5]
.. .. | |--leaf "Georgia"
.. .. | `--leaf "Tennessee"
.. .. `--[dendrogram w/ 2 branches and 3 members at h = 1.18,
midpoint = 0.75]
.. .. |--leaf "North Carolina"
.. .. `--[dendrogram w/ 2 branches and 2 members at h = 0.787,
midpoint = 0.5]
.. .. |--leaf "Mississippi"
.. .. `--leaf "South Carolina"
> #Plot the top part (high-level cluster structure)
```

```
> plot(cut_tree$upper, main = "Upper Dendrogram (Top Cluster Structure)")
```

Upper Dendrogram (Top Cluster Structure)



```
> #Plot one of the resulting clusters (say, 2nd)
> plot(cut_tree$lower[[2]], main="Detailed View of Cluster 2")
```

Detailed View of Cluster 2

